

SOC-5811 Week 12: Assumptions and Diagnostics

Nick Graetz

University of Minnesota, Department of Sociology

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1/54



REVIEW: FIXED EFFECTS

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 - ▶ If we control for unit with fixed effects (or by manually demeaning), we are treating it as a constant, non-random variable.
- ▶ When is a regression a “fixed effects model?”
 - ▶ When we include *at least* a fixed effect for unit.
 - ▶ We can also include lots of other predictors as fixed effects; this is convenient if they have many levels and are simply control variables.

ASSUMPTIONS OF REGRESSION

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- ▶ The model should include all relevant predictors.
- ▶ The model should generalize to the cases in which it will be applied.

ASSUMPTIONS OF REGRESSION

(2) Representativeness:

- ▶ A regression model is fit to data and is used to make inferences about a larger population, hence the implicit assumption in interpreting regression coefficients is that the sample is representative of the population.



ASSUMPTIONS OF REGRESSION

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- ▶ If linearity is violated, perhaps a predictor should be put in as $1/x$ or $\log(x)$ instead of simply linearly.
- ▶ More on this in **Week 14**.



ASSUMPTIONS OF REGRESSION

(4) Independence of errors:

- ▶ The simple regression model assumes that the errors from the prediction line are independent, an assumption that is violated in time series, spatial, and multilevel settings.



ASSUMPTIONS OF REGRESSION

(5) Equal variance of errors:

- ▶ Unequal error variance (also called heteroscedasticity, in contrast to equal variances, or homoscedasticity) can be an issue when a regression is used for probabilistic prediction.



ASSUMPTIONS OF REGRESSION

(6) Normality of errors:

- ▶ The distribution of the error term is relevant when predicting individual data points. For the purpose of estimating the regression line (as compared to predicting individual data points), the assumption of normality is typically barely important at all.



ASSUMPTIONS OF REGRESSION

(6) Normality of errors:

- ▶ The distribution of the error term is relevant when predicting individual data points. For the purpose of estimating the regression line (as compared to predicting individual data points), the assumption of normality is typically barely important at all.
- ▶ The regression model does *not* assume or require that predictors be normally distributed.



EXAMPLE

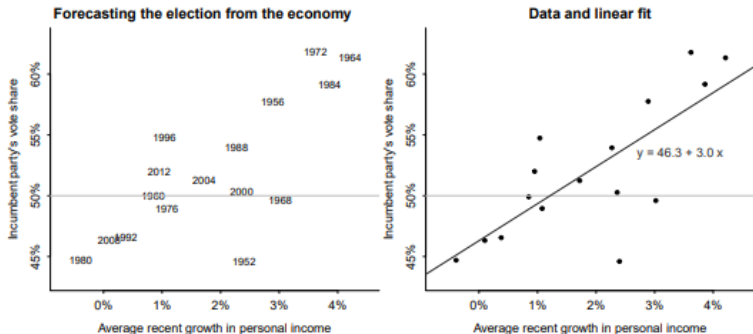


Figure 7.2 (a) Predicting elections from the economy: the data in Figure 7.1 expressed as a scatterplot, with one data point for each year, (b) the data with the linear fit, $y = 46.3 + 3.0x$. Repeated from Figure 1.1.

EXAMPLE

- ▶ Validity
- ▶ Representativeness
- ▶ Additivity and linearity
- ▶ Independence of errors
- ▶ Equal variance of errors
- ▶ Normality of errors



VIOLATIONS OF ASSUMPTIONS

Post-hoc solutions:

- ▶ Validity: measurement error models.



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VIOLATIONS OF ASSUMPTIONS

Post-hoc solutions:

- ▶ Validity: measurement error models.
- ▶ Representativeness: selection models.
- ▶ Linearity: nonadditive and nonlinear models.
- ▶ Independence of errors: correlated errors or latent variables.



VIOLATIONS OF ASSUMPTIONS

Change the data/model:

- ▶ Obtaining cleaner data.

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Change the data/model:

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- ▶ Adding interactions.
- ▶ Transforming predictors.

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Change the question/interpretation:

- ▶ Change the research question to align to the data.



VIOLATIONS OF ASSUMPTIONS

Change the question/interpretation:

- ▶ Change the research question to align to the data.
- ▶ Stick to descriptive interpretation that is less causal and does not extrapolate.



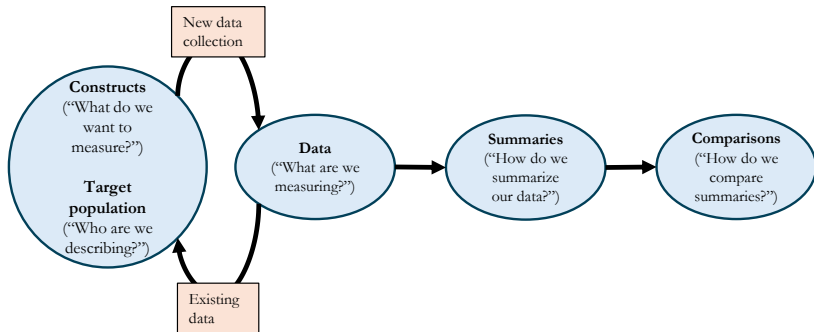
VIOLATIONS OF ASSUMPTIONS

Change the question/interpretation:

- ▶ Change the research question to align to the data.
- ▶ Stick to descriptive interpretation that is less causal and does not extrapolate.
- ▶ Defining the population to match the sample.



► Epistemology



VIOLATIONS OF ASSUMPTIONS

- ▶ How do we check for violations of our assumptions?

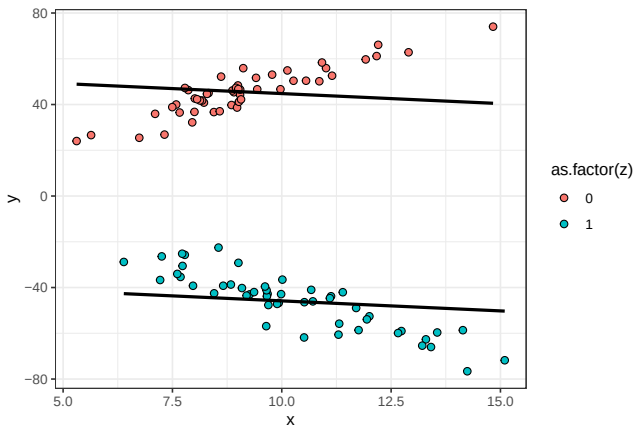


VIOLATIONS OF ASSUMPTIONS

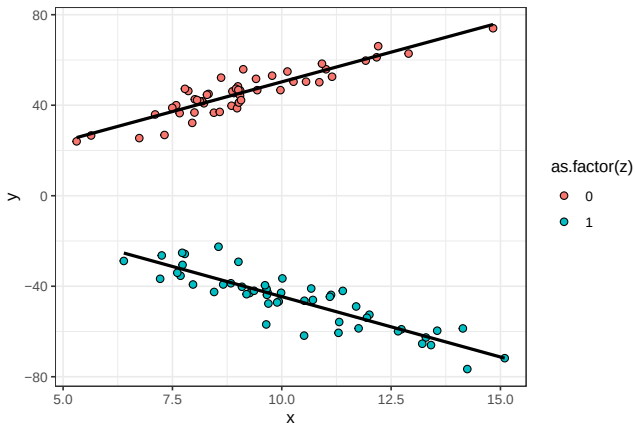
- ▶ How do we check for violations of our assumptions?
- ▶ **Plots.**

VIOLATIONS OF ASSUMPTIONS

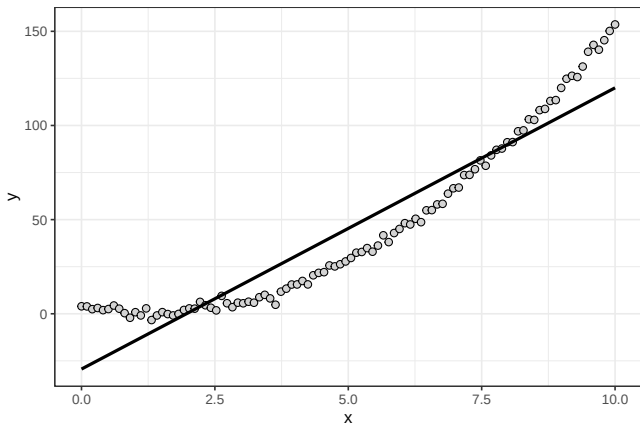
► Interactions



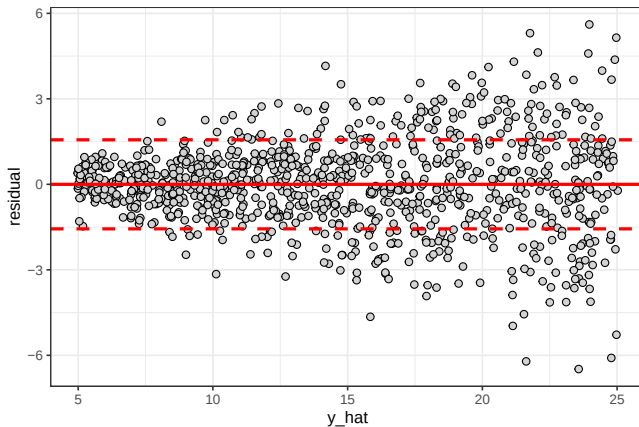
VIOLATIONS OF ASSUMPTIONS



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VIOLATIONS OF ASSUMPTIONS



MODEL COMPARISON

- ▶ If we want to choose the “best” model specification that relaxes our assumptions above (e.g., transforming a variable), how would we do that?



MODEL COMPARISON

$$R^2 = 1 - \left(\frac{\hat{\sigma}^2}{s_y^2} \right)$$

- We can only compare models fit to the same data; we cannot compare models fit to different data.

MODEL COMPARISON

$$R^2 = 1 - \left(\frac{\hat{\sigma}^2}{s_y^2} \right)$$

- ▶ We can only compare models fit to the same data; we cannot compare models fit to different data.
- ▶ How would I achieve an R^2 of 1 (i.e., perfect prediction)?

MODEL COMPARISON

- ▶ Remember connecting every dot? We could design a regression model that makes these predictions. Why do we not do that?



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- ▶ **Overfitting:** the model is not generalizable outside our particular sample.



MODEL COMPARISON

- ▶ Remember connecting every dot? We could design a regression model that makes these predictions. Why do we not do that?
- ▶ **Overfitting:** the model is not generalizable outside our particular sample.
- ▶ How do I know if I'm overfitting?



CROSS-VALIDATION

- ▶ **External validation**



CROSS-VALIDATION

- ▶ **External validation**
- ▶ Apply my fitted model to a new dataset measuring the same thing.



CROSS-VALIDATION

- ▶ **External validation**
- ▶ Apply my fitted model to a new dataset measuring the same thing.
- ▶ Cross-validation!

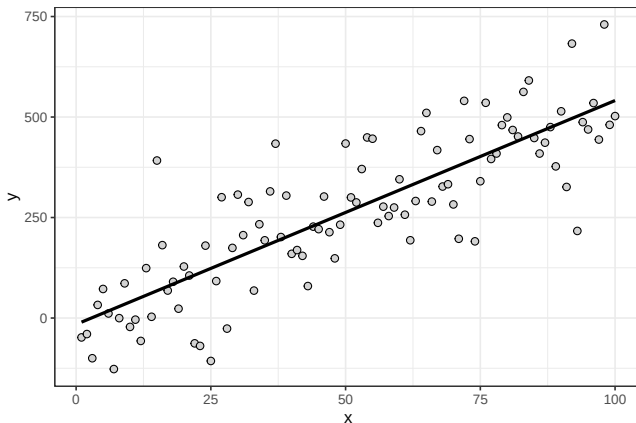


REVIEW

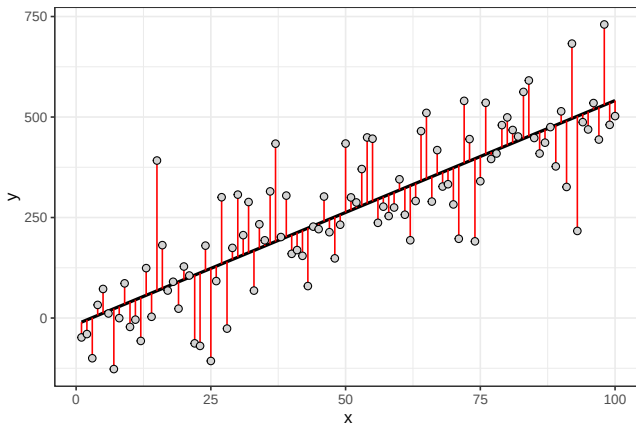
- Pick up here on Wednesday.



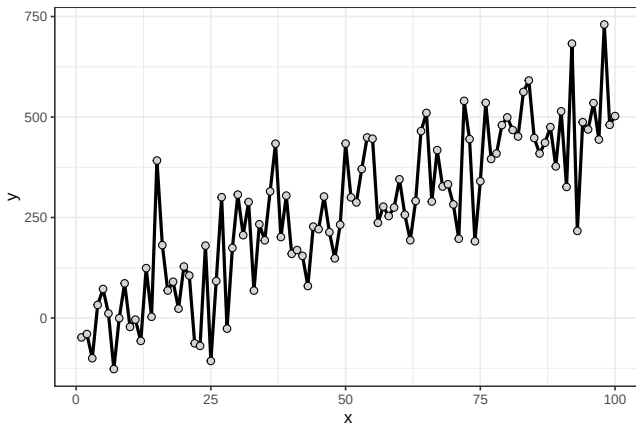
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- ▶ How do we compare the fit of these models? Why can't we just see which model has a higher R^2 , the proportion of variance in y predicted by each model?
- ▶ R^2 only characterizes how well the model predicts *in-sample* variation, but we really care about how well the model can predict *out-of-sample* variation.



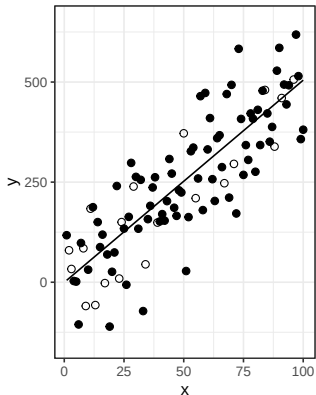
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- ▶ R^2 only characterizes how well the model predicts *in-sample* variation, but we really care about how well the model can predict *out-of-sample* variation.
- ▶ What is “out-of-sample” variation? Variation in y that we did not observe!

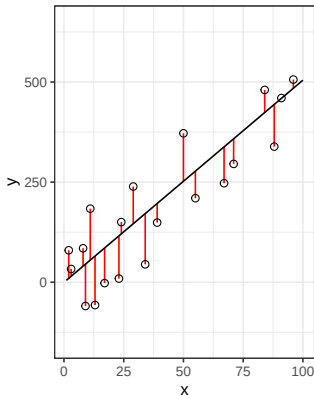


CROSS-VALIDATION

Step 1: Train

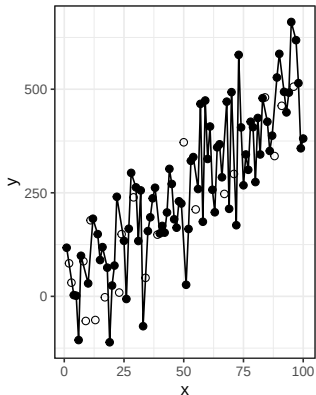


Step 2: Test

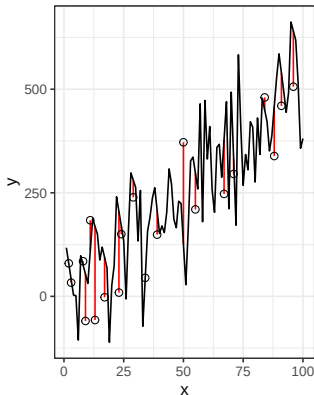


CROSS-VALIDATION

Step 1: Train



Step 2: Test



CROSS-VALIDATION

Out-of-sample root mean squared error (OOS RMSE):

- ▶ **OLS:** OOS RMSE = 84.36
- ▶ **Overfit:** OOS RMSE = 111.54



CROSS-VALIDATION

- ▶ Model comparison is always important, but it is particularly important when we care about *prediction*.

CROSS-VALIDATION

- ▶ You work as a data scientist at Netflix and have been tasked with predicting what users will watch next. You have access to all users demographics and watch histories.



CROSS-VALIDATION

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1. Randomly split data into **training data** and **test data**.
 2. Fit various models to **training data** (basic linear model, generalized linear models, machine learning models: gradient boosting machine, Bayesian additive regression trees, random forest).
 3. Use models to make predictions for **test data**.
 4. Compare fit statistics (e.g., root mean squared error) to select the highest performing model.



CROSS-VALIDATION

- ▶ **What do we learn about the relationships between predictors and the outcome when we select the "best" predictive model using cross-validation?**



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 - ▶ Potentially nothing! But that isn't always our goal.



CROSS-VALIDATION

- ▶ **What do we learn about the relationships between predictors and the outcome when we select the "best" predictive model using cross-validation?**
 - ▶ Potentially nothing! But that isn't always our goal.
- ▶ **Why does one model perform better than another? Does it matter?**



MULTICOLLINEARITY

- ▶ What is it?



MULTICOLLINEARITY

- ▶ What is it?
- ▶ How do I know?



MULTICOLLINEARITY

- ▶ What is it?
- ▶ How do I know?
- ▶ When is this a problem?



MULTICOLLINEARITY

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MULTICOLLINEARITY

- ▶ What is it?
- ▶ How do I know?
- ▶ When is this a problem?
- ▶ What should I do?
 - ▶ Do nothing.



MULTICOLLINEARITY

- ▶ What is it?
- ▶ How do I know?
- ▶ When is this a problem?
- ▶ What should I do?
 - ▶ Do nothing.
 - ▶ Drop variables.



MULTICOLLINEARITY

- ▶ What is it?
- ▶ How do I know?
- ▶ When is this a problem?
- ▶ What should I do?
 - ▶ Do nothing.
 - ▶ Drop variables.
 - ▶ Combine variables (e.g., principal components analysis).

INTERACTIONS AND TRANSFORMATIONS

- ▶ Validity
- ▶ Representativeness
- ▶ Additivity and linearity
- ▶ Independence of errors
- ▶ Equal variance of errors
- ▶ Normality of errors



INTERACTIONS AND TRANSFORMATIONS

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INTERACTIONS AND TRANSFORMATIONS

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- ▶ Representativeness
- ▶ **Additivity and linearity**
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INTERACTIONS AND TRANSFORMATIONS

- ▶ How do we relax our assumptions around additivity and linearity?



INTERACTIONS AND TRANSFORMATIONS

- Additivity: add interaction terms.

Assume additivity across treatment and confounders:

$$y_i = X_i + Z_i + W_i$$

Allow for treatment interaction:

$$y_i = X_i + Z_i + (X_i \times Z_i) + W_i$$

Allow for confounder interaction:

$$y_i = X_i + Z_i + W_i + (Z_i \times W_i)$$

INTERACTIONS AND TRANSFORMATIONS

- ▶ Additivity: add interaction terms.
- ▶ Let's assume we are interested in the causal effect of X on Y, and Z and W are confounders.

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INTERACTIONS AND TRANSFORMATIONS

- How do we decide whether to include an interaction?

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- ▶ Linearity: transform variables.
- ▶ We aren't talking about *shifting* or *rescaling* predictor variables, which changes our interpretation of the coefficients on those variables.
- ▶ We are talking about changing the *functional relationship* between a predictor variable and the outcome.

INTERACTIONS AND TRANSFORMATIONS

- Log-transforming



INTERACTIONS AND TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials

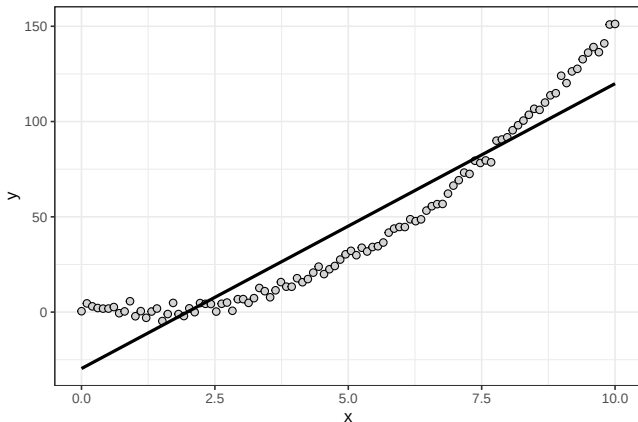
INTERACTIONS AND TRANSFORMATIONS

- ▶ Log-transforming
- ▶ Polynomials
- ▶ Splines



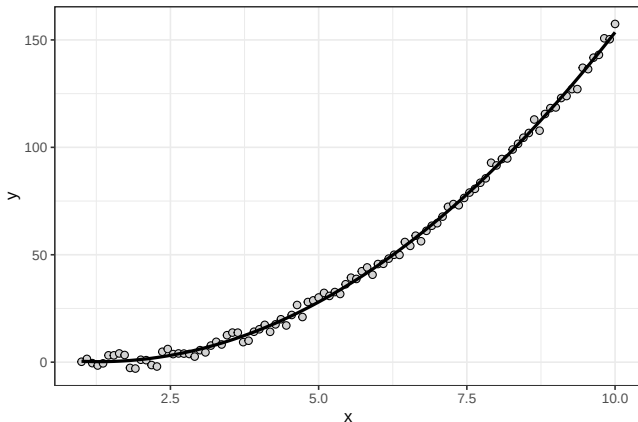
INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 X_i$$



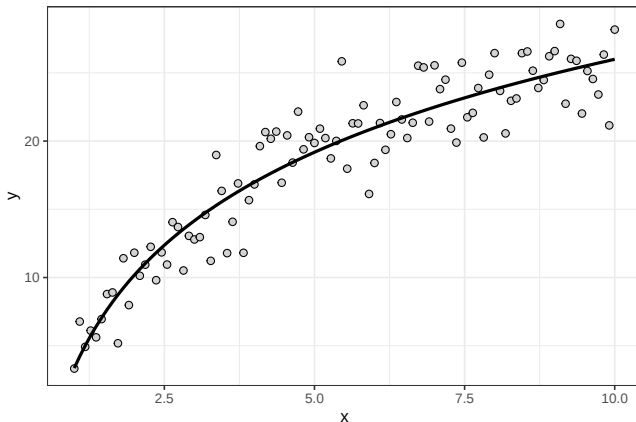
INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2$$



INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$



INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

`summary(mod)`

```
##
## Call:
## lm(formula = y ~ log_x, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7546 -1.3002 -0.0268  1.3241  5.8073
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.3639     0.5904   5.698 1.27e-07 ***
## log_x         9.8281     0.3545  27.722 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.119 on 98 degrees of freedom
## Multiple R-squared:  0.8869, Adjusted R-squared:  0.8857
## F-statistic: 768.5 on 1 and 98 DF,  p-value: < 2.2e-16
```

$$\beta_1 = 10.6$$

INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

$$\beta_1 = 10.6$$

Interpretation:

- ▶ A one-unit increase in $\ln(X)$ is associated with a 10.6 increase in Y .

INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1 \log(X_i)$$

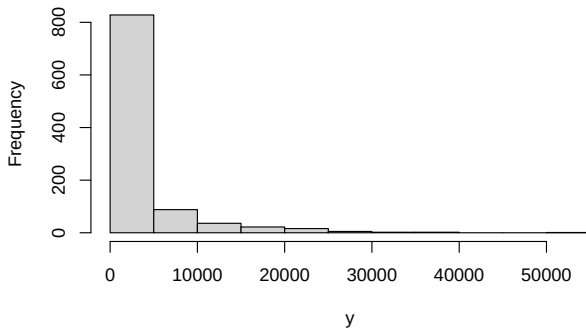
$$\beta_1 = 10.6$$

Interpretation:

- ▶ A one-unit increase in $\ln(X)$ is associated with a 10.6 increase in Y .
- ▶ A 1% increase in X is associated with a .106 increase in Y .

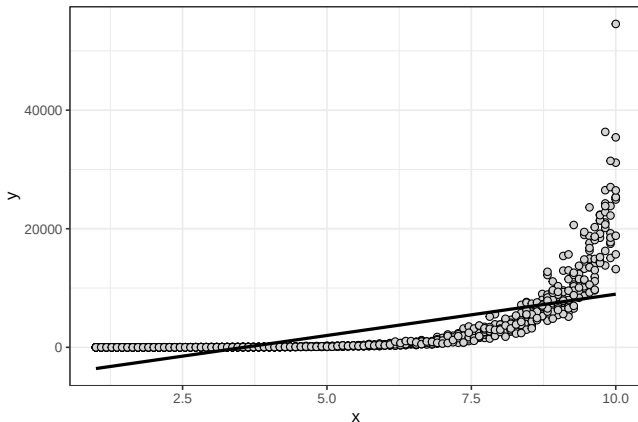
INTERACTIONS AND TRANSFORMATIONS

Histogram of y



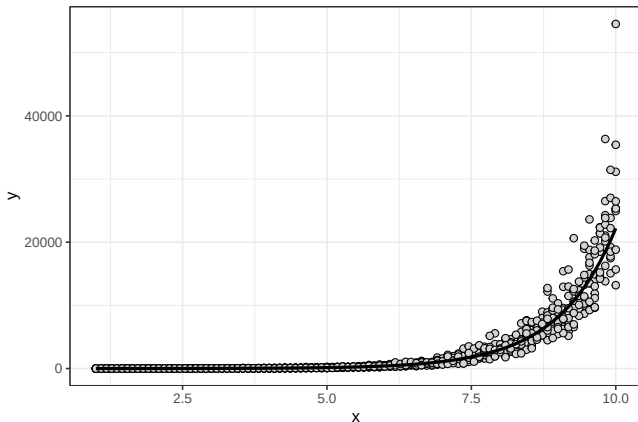
INTERACTIONS AND TRANSFORMATIONS

$$y_i = \beta_0 + \beta_1(X_i)$$



INTERACTIONS AND TRANSFORMATIONS

$$\log(y_i) = \beta_0 + \beta_1(X_i)$$



INTERACTIONS AND TRANSFORMATIONS

- Why would we do this?



INTERACTIONS AND TRANSFORMATIONS

- ▶ Why would we do this?
- ▶ Avoid negative predictions for an continuous outcome variable that is bounded at 0.



INTERACTIONS AND TRANSFORMATIONS

- ▶ Why would we do this?
- ▶ Avoid negative predictions for an continuous outcome variable that is bounded at 0.
- ▶ If we think the relationship between X and Y is multiplicative rather than additive.

